Project 3: Technical Debt Remediation, Access Automation, & Operational Risk Matrix

💡 **Portfolio Methodology Note:** This project profile represents a data-driven risk intervention and process automation initiative that I independently identified, and championed. Backed by my PMP and Agile mindset, I exposed a critical operational single point of failure (SPOF), coordinated the cross-functional stakeholder alignment, and secured executive buy-in for a permanent system upgrade. While the underlying API-integrated spreadsheet tool were engineered by the Engineer, this document demonstrates my hands-on capability to identify systemic risk, define compliance requirements, and lead teams through digital optimization cycles.

📌 Executive Case Overview

- **Domain:** Technical Debt Remediation, Access Control, Operational Continuity, Risk Analysis
- **Target Framework Benchmarks:** NIST SP 800-53 Rev. 5 (PM-28: Risk Management Strategy, PL-4: Rules of Behavior, CM-9: Configuration Management Plan)
- **Objective:** Identify and eliminate a critical manual bottleneck within a high-priority federal pre-order validation environment, transforming a 23-step process dependent on tribal knowledge into a scalable, automated pipeline.

🛑 The Systemic Risks & Manual Bottlenecks

Upon stepping into the role of Account Advisor for a high-priority federal account, I conducted an operational audit of our data workflows and uncovered a severe structural vulnerability. The pre-order validation phase, a 23-step harness compliance check, was entirely manual, requiring roughly 60 minutes per order to prepare.

The immediate operational risks included:

1. **Critical Single Point of Failure (SPOF):** The institutional knowledge required to execute this 23-step validation lived exclusively with a single employee who had departed the team, leaving the remaining team unable to independently process requests.
2. **Total Operational Standstill:** The team inherited a mounting backlog of stalled orders for a high-priority partner, creating fulfillment delays that breached SLA..
3. **Managerial Resistance:** Leadership initially blocked modifications due to concerns over disrupting external customer workflows, forcing the department into a rigid and vulnerable posture.

🛠️ Risk Intervention & Control Mapping

1. Risk Management Strategy & Escalation (NIST SP 800-53: PM-28)
    - **The GRC Approach:** Identifying and formalizing hidden operational bottlenecks to secure senior leadership visibility and justify engineering resource allocation.
    - **The Execution:** Operating on my own initiative, I documented the timeline delays caused by the manual validation checks, including multi-month order stalls and offline logjams. I formally escalated this systemic risk to leadership, exposing the organizational liability and shifting the conversation from a speculative posture to a data-backed business case.

2. Cross-Functional Change Championship (NIST SP 800-53: PL-4, CM-9)
    - **The GRC Approach:** Organizing multi-disciplinary stakeholders to transition an enterprise from a high-risk people-dependent model to a repeatable process-dependent environment.
    - **The Execution:** I championed a permanent technical change, bringing together the required cross-functional divisions and looping in the Engineering team to audit the manual workflow. Simultaneously, I negotiated a temporary interim support bridge with the former process owner to clear the immediate backlog, protecting the partner experience while the technical team engineered the permanent automation tool.

3. Quantitative Risk Tracking Infrastructure (NIST SP 800-53: PM-28)
    - **The GRC Approach:** Designing data-capture mechanisms to convert abstract feature requirements into structured, quantified pipeline evidence.
    - **The Execution:** To address executive demands for explicit ROI justification, I designed and launched an internal Customer Demand Signal Tracker. This decentralized framework allowed the account team to systematically document client requests for non-compliant product lines, capturing account tiers, unit scales, and total projected contract values to build the department's first aggregated federal revenue map.

📈 Quantifiable Operational Multipliers

- **90% Processing Compression:** The intervention and the resulting tool built by Engineering automated the 23 manual checks, collapsing order validation time from roughly 60 minutes to under 1 minute per execution.

- **Revenue Protection:** Cleared the multi-agency fulfillment logjam, preventing further timeline slippage and securing high-priority federal account revenue.
- **Headcount Scalability:** Unlocked the department's capacity to handle bulk enterprise orders without requiring a linear increase in team headcount.
- **Standardized Operational Blueprint:** Formulated standard operating procedures and a five-point Lessons Learned Register, adopted as the corporate standard for future federal account management.

🗂️ Project Repository Artifacts

See artifacts.md for the sanitized data-capture frameworks and process-mapping tools used to execute this risk turnaround:

- Artifact A: Pre-Order Compliance Automation Requirements Blueprint
- Artifact B: Customer Demand Signal Tracker Data Schema